INCH-POUND

MIL-M-38510/755A
29 JUNE 1992
SUPERSEDING
MIL-M-38510/755
28 December 1989

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, ADVANCED CMOS,
TRANSCEIVERS, MONOLITHIC SILICON, POSITIVE LOGIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, advanced CMOS, logic microcircuits. Two product assurance classes and a choice of case outlines, lead finishes, and radiation hardness assurance (RHA) are provided and are reflected in the complete Part on Identifying Number (PIN). The PIN shall be formulated in accordance with MIL-M-38510.

1.2 Classification.

1.2.1 Device type. The device types shall be as follows:

Device type	Circuit
01	To be included after dating
02	To be included after dating
03	Octal bidirectional transceiver with three-state outputs
04	To be included after dating
05	Octal noninverting bus transceiver with three-state outputs
06	Octal bidirectional transceiver with three-state outputs
07	Octal bidirectional transceiver with three-state outputs
08	Octal transceiver/register with three-state outputs
09	Octal transceiver/register with three-state outputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
K	F-6 (24-lead, .640" x .420" x .090"), flat package
L	D-9 (24-lead, 1.280" x .310" x 200"), dual-in-line package
R	D-8 (20-lead, 1.060" x .310" x .200"), dual-in-line package
S	F-9 (20-lead, .540" x .300" x .100"), flat package
2	C-2 (20 terminal, .358" x .358" x .100"), square chip carrier package
3	C-4 (28 terminal, .460" x .460" x .100"), square chip carrier package

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Laboratory (RL/ERDS), Griffiss AFB, NY 13441-5700, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5962
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1.3 Absolute maximum ratings. 1/ 2/

Supply voltage range (V_{CC})	-0.5 V dc to +6.0 V dc
DC input voltage range (V_{IN})	-0.5 V dc to V_{CC} + 0.5 V dc
DC output voltage range (V_{OUT})	-0.5 V dc to V_{CC} + 0.5 V dc
Clamp diode current (I_{IK}, I_{OK})	±20 mA
DC output current (I_{OUT})	±50 mA
DC V_{CC} or GND current (I_{CC}, I_{GND})	±100 mA
Storage temperature range (t_{STG})	-65°C to +150°C
Maximum power dissipation (P_D)	500 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (Θ_{JC})	See MIL-M-38510, Appendix C
Junction temperature (T_J)	+175°C

1.4 Recommended operating conditions. 2/ 3/ 4/

Supply voltage range (V_{CC})	+3.0 V dc to +5.5 V dc
Input voltage range (V_{IN})	+0.0 V dc to V_{CC}
Output voltage range (V_{OUT})	+0.0 V dc to V_{CC}
Case operating temperature range (T_C)	-55°C to +125°C
Maximum low level input voltage (V_{IL})	0.90 V at V_{CC} = 3.0 V 1.35 V at V_{CC} = 4.5 V 1.65 V at V_{CC} = 5.5 V
Minimum high level input voltage (V_{IH})	2.10 V at V_{CC} = 3.0 V 3.15 V at V_{CC} = 4.5 V 3.85 V at V_{CC} = 5.5 V
Input rise and fall rate (t_r, t_f) maximum: V_{CC} = 3.6 V, V_{CC} = 5.5 V	8 ns/V

Minimum set-up time, high or low, bus to clock (t_s):

		Device types 08	09
V_{CC} = 3.0 V dc;	T_C = +25°C, -55°C	5.0 ns	3.0 ns
	T_C = +125°C	6.0 ns	4.0 ns
V_{CC} = 4.5 V dc;	T_C = +25°C, -55°C	4.0 ns	2.0 ns
	T_C = +125°C	4.5 ns	2.5 ns

Minimum hold time, high or low, bus to clock (t_h):

		Device types 08	09
V_{CC} = 3.0 V dc;	T_C = +25°C, +125°C	1.0 ns	0.0 ns
	T_C = -55°C	1.5 ns	0.0 ns
V_{CC} = 4.5 V dc;	T_C = +25°C, +125°C	1.5 ns	1.0 ns
	T_C = -55°C	2.0 ns	1.0 ns

Minimum clock pulse width (t_w), high or low:

		Device types 08	09
V_{CC} = 3.0 V dc;	T_C = +25°C, -55°C	5.0 ns	3.5 ns
	T_C = +125°C	5.0 ns	4.5 ns
V_{CC} = 4.5 V dc;	T_C = +25°C, -55°C	5.0 ns	2.0 ns
	T_C = +125°C	5.0 ns	3.5 ns

1/ Stresses above the absolute maximum rating may cause permanent damage to the device. Extended operation at the maximum levels may degrade performance and affect reliability. The maximum junction temperature may be exceeded for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883.

2/ Unless otherwise specified, all voltages are referenced to GND.

3/ Operation from 2.0 V dc to 3.0 V dc is provided for compatibility with data retention and battery back up systems. Data retention implies no input transitions and no stored data loss with the following conditions: $V_{IH} \geq$ 70 percent of V_{CC}, $V_{IL} \leq$ 30 percent of V_{CC}, $V_{OH} \geq$ 70 percent of V_{CC} at -20 µA, $V_{OL} \leq$ 30 percent of V_{CC} at 20 µA.

4/ Unless otherwise specified, the values listed above shall apply over the full V_{CC} and T_C recommended operating range.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONICS INDUSTRIES ASSOCIATION (EIA)

JEDEC Standard No. 17 - A Standardized Test Procedure for the Characterization of LATCH-UP in CMOS Integrated Circuits.

JEDEC Standard No. 20 - Standard for Description of 54/74ACXXXXX and 54/74ACTXXXXX Advanced High-Speed CMOS Devices.

(Applications for copies should be addressed to the Electronics Industries Association, 2001 Eye Street, N.W., Washington, DC 20006.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.1.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.1.2 Truth tables. The truth tables shall be as specified on figure 2.

3.1.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

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3.2 Electrical performance characteristics and postirradiation parameter limits. Unless otherwise specified, the electrical performance characteristics, and postirradiation parameter limits are as specified in table I. Electrical performance characteristics apply over the case operating temperature range specified. Test conditions for these specified characteristics and limits are as specified in table I. A pin for pin conditions and testing sequence for table I parameters shall be maintained and available upon request from the qualifying activity, on qualified devices.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table I. Radiation hardness assurance level designators M, D, and R (see MIL-M-38510) in table I are postirradiation end-point electrical parameters.

3.5 Correctness of indexing and marking. All devices shall be subjected to the final electrical tests specified in table II after PIN marking (in accordance with MIL-M-38510) to verify that they are correctly indexed and identified by PIN. Optionally, an approved electrical test may be devised especially for this requirement.

3.5.1 Radiation hardness assurance identifier. The radiation hardness assurance identifier shall be in accordance with MIL-M-38510 and herein (see 3.4).

3.6 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 37 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.1.1 Burn-in and life test circuits. Burn-in and life test circuits shall be constructed so that the devices are stressed at the maximum operating conditions stated in 4.2b or 4.2c as applicable or equivalent as approved by the qualifying activity.

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TABLE I. Electrical performance characteristics.

Test and MIL-STD-883 test method	Symbol	Test conditions 1/ $-55°C \le T_C \le +125°C$ $3.0\ V \le V_{CC} \le 5.5\ V$ unless otherwise specified	Device type 2/	V_{CC}	Group A subgroups	Limits 1/ Min	Limits 1/ Max	Unit
High level output voltage 3006	V_{OH1} 3/	For all inputs affecting output under test $V_{IN} = V_{IH}$ or V_{IL} $V_{IH} = 2.10\ V$ $V_{IL} = 0.90\ V$ For all other inputs $V_{IN} = V_{CC}$ or GND $I_{OH} = -50\ \mu A$	All	3.0 V	1,2,3	2.9		V
	V_{OH2} 3/	For all inputs affecting output under test $V_{IN} = V_{IH}$ or V_{IL} $V_{IH} = 3.15\ V$ $V_{IL} = 1.35\ V$ For all other inputs $V_{IN} = V_{CC}$ or GND $I_{OH} = -50\ \mu A$	All	4.5 V	1,2,3	4.4		
	V_{OH3} 4/ 5/	For all inputs affecting output under test $V_{IN} = V_{IH}$ or V_{IL} $V_{IH} = 3.85\ V$ $V_{IL} = 1.65\ V$ For all other inputs $V_{IN} = V_{CC}$ or GND $I_{OH} = -50\ \mu A$	All	5.5 V	1,2,3	5.4		
			M, D, R					
			03		1	5.4		
						5.4		
						5.4		
	V_{OH4} 3/	For all inputs affecting output under test $V_{IN} = V_{IH}$ or V_{IL} $V_{IH} = 2.10\ V$ $V_{IL} = 0.90\ V$ For all other inputs $V_{IN} = V_{CC}$ or GND $I_{OH} = -4\ mA$	All	3.0 V	1,2,3	2.4		
	V_{OH5} 4/ 5/	For all inputs affecting output under test $V_{IN} = V_{IH}$ or V_{IL} $V_{IH} = 3.15\ V$ $V_{IL} = 1.35\ V$ For all other inputs $V_{IN} = V_{CC}$ or GND $I_{OH} = -24\ mA$	All	4.5 V	1,2,3	3.7		
			M, D, R					
			03		1	3.7		
						3.7		
						3.7		

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test and MIL-STD-883 test method	Symbol	Test conditions 1/ $-55°C \le T_C \le +125°C$ $3.0\ V \le V_{CC} \le 5.5\ V$ unless otherwise specified		Device type 2/	V_{CC}	Group A subgroups	Limits 1/		Unit
							Min	Max	
High level output voltage 3006	V_{OH6} 3/	For all inputs affecting output under test $V_{IN} = V_{IH}$ or V_{IL} $V_{IH} = 3.85\ V$ $V_{IL} = 1.65\ V$ For all other inputs $V_{IN} = V_{CC}$ or GND $I_{OH} = -24\ mA$		All	5.5 V	1,2,3	4.7		V
	V_{OH7} 4/ 5/ 6/	For all inputs affecting output under test $V_{IN} = V_{IH}$ or V_{IL} $V_{IH} = 3.85\ V$ $V_{IL} = 1.65\ V$ For all other inputs $V_{IN} = V_{CC}$ or GND $I_{OH} = -50\ mA$	M D R	All 03	5.5 V	1,2,3 1	3.85 3.85 3.85		
Low level output voltage 3007	V_{OL1} 3/	For all inputs affecting output under test $V_{IN} = V_{IH}$ or V_{IL} $V_{IH} = 2.10\ V$ $V_{IL} = 0.90\ V$ For all other inputs $V_{IN} = V_{CC}$ or GND $I_{OL} = 50\ \mu A$		All	3.0 V	1,2,3		0.1	V
	V_{OL2} 3/	For all inputs affecting output under test $V_{IN} = V_{IH}$ or V_{IL} $V_{IH} = 3.15\ V$ $V_{IL} = 1.35\ V$ For all other inputs $V_{IN} = V_{CC}$ or GND $I_{OL} = 50\ \mu A$		All	4.5 V	1,2,3		0.1	
	V_{OL3} 4/ 5/	For all inputs affecting output under test $V_{IN} = V_{IH}$ or V_{IL} $V_{IH} = 3.85\ V$ $V_{IL} = 1.65\ V$ For all other inputs $V_{IN} = V_{CC}$ or GND $I_{OL} = 50\ \mu A$	M D R	All 03	5.5 V	1,2,3 1		0.1 0.1 0.1	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test and MIL-STD-883 test method	Symbol	Test conditions 1/ $-55°C \le T_C \le +125°C$ $3.0\ V \le V_{CC} \le 5.5\ V$ unless otherwise specified		Device type 2/	V_{CC}	Group A subgroups	Limits 1/		Unit
							Min	Max	
Low level output voltage 3007	V_{OL4} 3/	For all inputs affecting output under test $V_{IN} = V_{IH}$ or V_{IL} $V_{IH} = 2.10\ V$ $V_{IL} = 0.90\ V$ For all other inputs $V_{IN} = V_{CC}$ or GND $I_{OL} = 12\ mA$		All	3.0 V	1,3		0.4	V
						2		0.5	
	V_{OL5} 4/ 5/	For all inputs affecting output under test $V_{IN} = V_{IH}$ or V_{IL} $V_{IH} = 3.15\ V$ $V_{IL} = 1.35\ V$ For all other inputs $V_{IN} = V_{CC}$ or GND $I_{OL} = 24\ mA$		All	4.5 V	1,3		0.4	
						2		0.5	
			M	03		1		0.4	
			D					0.4	
			R					0.4	
	V_{OL6} 3/	For all inputs affecting output under test $V_{IN} = V_{IH}$ or V_{IL} $V_{IH} = 3.85\ V$ $V_{IL} = 1.65\ V$ For all other inputs $V_{IN} = V_{CC}$ or GND $I_{OL} = 24\ mA$		All	5.5 V	1,3		0.4	
						2		0.5	
	V_{OL7} 4/ 5/ 6/	For all inputs affecting output under test $V_{IN} = V_{IH}$ or V_{IL} $V_{IH} = 3.85\ V$ $V_{IL} = 1.65\ V$ For all other inputs $V_{IN} = V_{CC}$ or GND $I_{OL} = 50\ mA$		All	5.5 V	1,2,3		1.65	
			M	03		1		1.65	
			D					1.65	
			R					1.65	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test and MIL-STD-883 test method	Symbol	Test conditions 1/ $-55°C \le T_C \le +125°C$ 3.0 V ≤ V_{CC} ≤ 5.5 V unless otherwise specified		Device type 2/	V_{CC}	Group A subgroups	Limits 1/ Min	Limits 1/ Max	Unit
Positive input clamp voltage 3022	V_{IC+} 4/ 5/	V_{CC} = GND For input under test I_{IN} = 1 mA		All		1	0.4	1.5	V
			M	03		1	0.4	1.5	
			D				0.4	1.5	
			R				0.4	1.5	
Negative input clamp voltage 3022	V_{IC-} 4/ 5/	V_{CC} = Open For input under test I_{IN} = -1 mA		All		1	-0.4	-1.5	V
			M	03		1	-0.4	-1.5	
			D				-0.4	-1.5	
			R				-0.4	-1.5	
Input current low 3009	I_{IL} 4/ 5/	For input under test V_{IN} = GND For all other inputs V_{IN} = V_{CC} or GND		All	5.5 V	1		-0.1	µA
						2		-1.0	
			M	03				-0.1	
			D			1		-0.1	
			R					-0.1	
Input current high 3010	I_{IH} 4/ 5/	For input under test V_{IN} = V_{CC} For all other inputs V_{IN} = V_{CC} or GND		All	5.5 V	1		0.1	µA
						2		1.0	
			M	03				0.1	
			D			1		0.1	
			R					0.1	
Input/output capacitance 3012	$C_{I/O}$ 8/	See 4.4.1c T_C = +25°C		All	5.5 V	4		15	pF
Input capacitance (control inputs) 3012	C_{IN}	See 4.4.1c T_C = +25°C		All	GND	4		10	pF
Power dissipation capacitance	C_{PD} 7/	See 4.4.1c T_C = +25°C		03	5.0 V	4		60	pF
				05,06,07				45	
				08				80	
				09				65	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test and MIL-STD-883 test method	Symbol	Test conditions 1/ $-55°C \le T_C \le +125°C$ 3.0 V $\le V_{CC} \le$ 5.5 V unless otherwise specified		Device type 2/	V_{CC}	Group A subgroups	Limits 1/		Unit
							Min	Max	
Supply current quiescent output three-state 3005	I_{CCZ} 4/ 5/ 8/	$V_{IN} = V_{CC}$ or GND		All	5.5 V	1		2.0	µA
						2		40.0	
			M	03		1		15	
			D					75	
			R					700	
Quiescent supply current output high 3005	I_{CCH} 4/ 5/	For all inputs $V_{IN} = V_{CC}$ or GND		All	5.5 V	1		2.0	µA
						2		40.0	
			M	03		1		15	
			D					75	
			R					700	
Quiescent supply current output low 3005	I_{CCL} 4/ 5/	For all other inputs $V_{IN} = V_{CC}$ or GND		All	5.5 V	1		2.0	µA
						2		40.0	
			M	03		1		15	
			D					75	
			R					700	
Three-state output leakage current low 3020	I_{OZL} 4/ 5/ 9/	V_{OUT} = GND		All	5.5 V	1		-0.6	µA
						2		-11.0	
			M	03		1		-1.0	
			D					-3.0	
			R					-20.0	
Three-state output leakage current high 3021	I_{OZH} 4/ 5/ 9/	V_{OUT} = VCC		All	5.5 V	1		0.6	µA
						2		11.0	
			M	03		1		1.0	
			D					3.0	
			R					20.0	
Low level ground bounce noise	V_{GBL} 10/ 11/	V_{LD} = 2.5 V, I_{OL} = +24 mA, V_I = 4.5 V or 0.0 V See figure 3		01,02	4.5 V	4	0.0	1000	mV
				03-09			0.0	2000	
High level ground bounce noise	V_{GBH} 10/ 11/	V_{LD} = 2.5 V, I_{OH} = -24 mA, V_I = 4.5 V or 0.0 V See figure 3		01,02	4.5 V	4	0.0	1000	mV
				03-09			0.0	2000	

See footnotes at end of table.

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Test and MIL-STD-883 test method	Symbol	Test conditions 1/ $-55°C \le T_C \le +125°C$ $3.0\ V \le V_{CC} \le 5.5\ V$ unless otherwise specified		Device type 2/	V_{CC}	Group A subgroups	Limits 1/ Min	Limits 1/ Max	Unit
Latch-up input/ output over-voltage	I_{CC} (O/V1) 12/	$t_w \ge 100\ \mu s$ $t_{cool} \ge t_w$ $5\ \mu s \le t_r \le 5\ ms$ $5\ \mu s \le t_f \le 5\ ms$ $V_{test} = 6.0\ V$ $V_{CCQ} = 5.5\ V$ $V_{over} = 10.5\ V$		All	5.5 V	2		200	mA
Latch-up input/ output positive over-current	I_{CC} (O/I1+) 12/	$t_w \ge 100\ \mu s$ $t_{cool} \ge t_w$ $5\ \mu s \le t_r \le 5\ ms$ $5\ \mu s \le t_f \le 5\ ms$ $V_{test} = 6.0\ V$ $V_{CCQ} = 5.5\ V$ $I_{trigger} = +120\ mA$		All	5.5 V	2		200	mA
Latch-up input/ output negative over-current	I_{CC} (O/I1-) 12/	$t_w \ge 100\ \mu s$ $t_{cool} \ge t_w$ $5\ \mu s \le t_r \le 5\ ms$ $5\ \mu s \le t_f \le 5\ ms$ $V_{test} = 6.0\ V$ $V_{CCQ} = 5.5\ V$ $I_{trigger} = -120\ mA$		All	5.5 V	2		200	mA
Latch-up supply over-voltage	I_{CC} (O/V2) 12/	$t_w \ge 100\ \mu s$ $t_{cool} \ge t_w$ $5\ \mu s \le t_r \le 5\ ms$ $5\ \mu s \le t_f \le 5\ ms$ $V_{test} = 6.0\ V$ $V_{CCQ} = 5.5\ V$ $V_{over} = 9.0\ V$		All	5.5 V	2		100	mA
Truth table test output voltage 3014	4/ 5/ 13/	$V_{IL} = 0.45\ V$ $V_{IH} = 2.50\ V$ Verify output V_O	M D R	03	3.0 V	7	L L L	H H H	
		$V_{IL} = 0.60\ V$ $V_{IH} = 3.70\ V$ Verify output V_O		All	4.5 V	7,8	L	H	
Maximum clock frequency 3003	f_{MAX}	$C_L = 50\ pF$, $R_L = 500\Omega$, See 4.4.1f		08,09	3.0 V	9,10,11	90		MHz
				08,09	4.5 V	9,10,11	100		

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

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Propagation delay time, data to output 3003	t_{PHL1}, t_{PLH1} 4/ 5/ 14/ 15/	C_L = 50 pF minimum, R_L = 500Ω, See figure 4		03	3.0 V	9,11	1.0	8.5	ns
						10	1.0	11.5	
				05		9,11	1.0	12.0	
						10	1.0	13.0	
				06		9,11	1.0	5.5	
						10	1.0	6.0	
				07		9,11	1.0	5.0	
						10	1.0	5.5	
				08,09		9,11	1.0	12.0	
						10	1.0	15.0	
			M	03		9	1.0	8.5	
			D				1.0	8.5	
			R				1.0	8.5	
		C_L = 50 pF minimum, R_L = 500Ω, See figure 4		03	4.5 V	9,11	1.0	6.5	ns
						10	1.0	8.5	
				05		9,11	1.0	8.5	
						10	1.0	9.5	
				06		9,11	1.0	4.0	
						10	1.0	4.5	
				07		9,11	1.0	4.0	
						10	1.0	4.5	
				08,09		9,11	1.0	8.0	
						10	1.0	10.0	
			M	03		9	1.0	6.5	
			D				1.0	6.5	
			R				1.0	6.5	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test and MIL-STD-883 test method	Symbol	Test conditions 1/ $-55°C \le T_C \le +125°C$ 3.0 V $\le V_{CC} \le$ 5.5 V unless otherwise specified	Device type 2/	V_{CC}	Group A subgroups	Limits 1/ Min	Limits 1/ Max	Unit
Propagation delay time, clock to bus 3003	t_{PHL2}, t_{PLH2} 14/ 15/	C_L = 50 pF minimum, R_L = 500Ω, See figure 4	08,09	3.0 V	9,11	1.0	16.0	ns
					10	1.0	20.0	
		C_L = 50 pF minimum, R_L = 500Ω, See figure 4	08,09	4.5 V	9,11	1.0	11.0	ns
					10	1.0	14.0	
Propagation delay time, SBA/SAB to bus 3003	t_{PHL3}, t_{PLH3} 14/ 15/	C_L = 50 pF minimum, R_L = 500Ω, See figure 4	08,09	3.0 V	9,11	1.0	13.5	ns
					10	1.0	17.0	
		C_L = 50 pF minimum, R_L = 500Ω, See figure 4	08,09	4.5 V	9,11	1.0	10.0	ns
					10	1.0	12.0	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test and MIL-STD-883 test method	Symbol	Test conditions 1/ $-55°C \le T_C \le +125°C$ $3.0\ V \le V_{CC} \le 5.5\ V$ unless otherwise specified		Device type 2/	V_{CC}	Group A subgroups	Limits 1/		Unit
							Min	Max	
Output enable time, OE, G OEAB, OEAB to output 3003	t_{PZH1}, t_{PZL1} 4/ 5/ 14/ 15/	C_L = 50 pF minimum, R_L = 500Ω, See figure 4		03	3.0 V	9,11	1.0	12.0	ns
						10	1.0	14.5	
				05		9,11	1.0	18.0	
						10	1.0	20.0	
				06		9,11	1.0	8.0	
						10	1.0	8.5	
				07		9,11	1.0	8.0	
						10	1.0	8.5	
				08,09		9,11	1.0	13.0	
						10	1.0	15.5	
			M	03			1.0	12.0	
			D				1.0	12.0	
			R				1.0	12.0	
		C_L = 50 pF minimum, R_L = 500Ω, See figure 4		03	4.5 V	9,11	1.0	9.0	ns
						10	1.0	10.5	
				05		9,11	1.0	12.0	
						10	1.0	13.0	
				06		9,11	1.0	6.0	
						10	1.0	6.5	
				07		9,11	1.0	6.0	
						10	1.0	6.5	
				08,09		9,11	1.0	9.5	
						10	1.0	11.0	
			M	03			1.0	9.0	
			D				1.0	9.0	
			R				1.0	9.0	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test and MIL-STD-883 test method	Symbol	Test conditions 1/ $-55°C \leq T_C \leq +125°C$ $3.0\ V \leq V_{CC} \leq 5.5\ V$ unless otherwise specified		Device type 2/	V_{CC}	Group A subgroups	Limits 1/		Unit
							Min	Max	
Output enable time, DIR to bus 3003	t_{PZH2}, t_{PZL2} 14/ 15/	C_L = 50 pF minimum, R_L = 500Ω, See figure 4		08,09	3.0 V	9,11	1.0	13.5	ns
						10	1.0	16.0	
		C_L = 50 pF minimum, R_L = 500Ω, See figure 4		08,09	4.5 V	9,11	1.0	10.5	ns
						10	1.0	12.5	
Output disable time, OE, G OEAB, OEAB to output 3003	t_{PHZ1}, t_{PLZ1} 4/ 5/ 14/ 15/	C_L = 50 pF minimum, R_L = 500Ω, See figure 4		03	3.0 V	9,11	1.0	12.0	ns
						10	1.0	14.0	
				05		9,11	1.0	15.0	
						10	1.0	16.5	
				06		9,11	1.0	7.5	
						10	1.0	8.0	
				07		9,11	1.0	7.5	
						10	1.0	8.0	
				08,09		9,11	1.0	12.0	
						10	1.0	14.0	
			M	03		9	1.0	12.0	
			D				1.0	12.0	
			R				1.0	12.0	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test and MIL-STD-883 test method	Symbol	Test conditions 1/ $-55°C \le T_C \le +125°C$ 3.0 V $\le V_{CC} \le$ 5.5 V unless otherwise specified		Device type 2/	V_{CC}	Group A subgroups	Limits 1/		Unit
							Min	Max	
Output disable time, OE, G OEAB, OEAB to output 3003	t_{PHZ1}, t_{PLZ1} 4/ 5/ 14/ 15/	C_L = 50 pF minimum, R_L = 500Ω, See figure 4		03	4.5 V	9,11	1.0	9.0	ns
						10	1.0	10.5	
				05		9,11	1.0	12.0	
						10	1.0	13.0	
				06		9,11	1.0	6.0	
						10	1.0	6.5	
				07		9,11	1.0	6.0	
						10	1.0	6.5	
				08,09		9,11	1.0	10.0	
						10	1.0	11.5	
			M	03		9	1.0	9.0	
			D				1.0	9.0	
			R				1.0	9.0	
Output disable time, DIR to bus 3003	t_{PHZ2}, t_{PLZ2} 14/ 15/	C_L = 50 pF minimum, R_L = 500Ω, See figure 4		08,09	3.0 V	9,11	1.0	14.0	ns
						10	1.0	16.5	
		C_L = 50 pF minimum, R_L = 500Ω, See figure 4		08,09	4.5 V	9,11	1.0	10.0	ns
						10	1.0	12.0	

1/ Each input/output, as applicable shall be tested at the specified temperature for the specified limits. Output terminals not designated shall be high level logic, low level logic, or open, except as follows:

a. V_{IC} (pos) tests, the GND terminal can be open. T_C = +25°C.

b. V_{IC} (neg) tests, the V_{CC} terminal shall be open. T_C = +25°C.

c. I_{CC} tests, the output terminal shall be open.

Additional detailed information on qualified devices (i.e. pin for pin conditions and testing (sequence) is available from the qualifying activity (DESC-EQM) upon request.

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TABLE I. Electrical performance characteristics - Continued.

2/ The word "All" in the device type column, means non-RHA and RHA preirradiation limits for all devices. Where M, D, and R in the conditions column are postirradiation limits for those device type specified in the device type column.

3/ This parameter is provided as design information only.

4/ RHA samples do not have to be tested at either -55°C or +125°C prior to irradiation.

5/ When performing postirradiation electrical measurements for any RHA level T_A = 25°C. Limits shown are guaranteed at T_A = 25°C ±5°C.

6/ Transmission driving tests are performed at V_{CC} = 5.5 V dc with a 2 ms duration maximum.

7/ Power dissipation capacitance (C_{PD}), determines the no load dynamic power consumption, $P_D = (C_{PD} + C_L)(V_{CC} \times V_{CC})f + (I_{CC} \times V_{CC})$, and the dynamic current consumption, $I_S = (C_{PD} + C_L) V_{CC} f + I_{CC}$. For both P_D and I_S, f is the frequency of the input signal.

8/ Set output enable control pins to V_{CC} or GND, as applicable, to disable the outputs.

9/ Three-state output conditions are required. For I_{OZL}, set outputs to high state. For I_{OZH}, set outputs to low state. Set output enable control pins to $V_{IL} = V_{IL(MAX)}$ and $V_{IH} = V_{IH(MIN)}$, as required. For I/O pins the I_{IH} and I_{IL} measurements shall not be directly performed. These measurements are included in the I_{OZH} and I_{OZL} limits, respectively.

10/ This test is for qualification only. Ground bounce tests are performed on a nonswitching (quiescent) output and are used to measure the magnitude of induced noise caused by other simultaneously switching outputs. The test is performed on a low noise bench test fixture with all outputs fully dc loaded (I_{OL} maximum and I_{OH} maximum = i.e, 24 mA) and 50 pF of load capacitance (figure 3). The loads must be located as close as possible to the device output. Inputs are then conditioned with 1 MHz pulse ($t_r = t_f$ = 3.5 ±1.5 ns) such that one output is forced low and all others (possible) switching simultaneously and in phase. The low level ground bounce noise is measured at the quiet output using a F.E.T. oscilloscope probe with at least 1 MΩ impedance. Measurement is taken from the peak of the largest positive pulse with respect to the nominal low level output voltage (figure 3). The device inputs are then conditioned such that the output under test is at a high nominal V_{OH} level. The high level ground bounce measurement is then measured from nominal V_{OH} level to the largest negative peak. This procedure is repeated such that all outputs are tested at a high and low level with a maximum number of outputs switching.

11/ When used in an asynchronous TTL compatible system, ground bounce (V_{GBL} and V_{GBH}) = 2000 mV can be a possible problem.

12/ See JEDEC standard 17 for electrically induced latch-up test methods and procedures. The values listed for $V_{trigger}$, $I_{trigger}$ and V_{over}, are to be accurate within ±5 percent.

13/ Tests shall be performed in sequence, attributes data only. Functional tests shall include the truth table and other logic patterns used for fault detection. Functional test shall be performed in sequence as approved by the qualifying activity on qualified devices. H ≥ 2.5 V, L < 2.5 V; high inputs = 3.7 V and low inputs = 0.6 V for V_{CC} = 4.5 V and H ≥ 1.5 V, L < 1.5 V; high inputs = 2.5 V and low inputs = 0.45 V for V_{CC} = 3.0 V. Tests at V_{CC} = 3.0 V are for RHA specified devices only (T_A = +25°C ±5°C). Functional tests at V_{CC} = 3.0 V are worst case for RHA specified devices.

14/ Devices are tested at V_{CC} = 3.0 V and V_{CC} = 4.5 V at T_C = 125°C for sample testing and at V_{CC} = 3.0 V and V_{CC} = 4.5 V at T_C = 25°C for screening. Other voltages of V_{CC} and temperatures are guaranteed, if not tested (see 4.4.1d).

15/ AC limits at V_{CC} = 5.5 V are equal to limits at V_{CC} = 4.5 V and guaranteed by testing at V_{CC} = 4.5 V. Minimum ac limits for V_{CC} = 5.5 V are 1 ns and guaranteed by guard banding V_{CC} = 4.5 V limits to 1.5 ns.

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Device type	03,06, 07	05	08, 09	
Case outline	R,S,2	R,S,2	K,L	3
Terminal number	Terminal symbol			
1	$T/\overline{R}$	OEAB	CAB	NC
2	A0	A0	SAB	CAB
3	A1	A1	DIR	SAB
4	A2	A2	A0	DIR
5	A3	A3	A1	A0
6	A4	A4	A2	A1
7	A5	A5	A3	A2
8	A6	A6	A4	NC
9	A7	A7	A5	A3
10	GND	GND	A6	A4
11	B7	B7	A7	A5
12	B6	B6	GND	A6
13	B5	B5	B7	A7
14	B4	B4	B6	GND
15	B3	B3	B5	NC
16	B2	B2	B4	B7
17	B1	B1	B3	B6
18	$\overline{B0}$	$\overline{B0}$	B2	B5
19	OE	$\overline{OEAB}$	B1	B4
20	V_{CC}	V_{CC}	$\overline{B0}$	B3
21	--	--	G	B2
22	--	--	SBA	NC
23	--	--	CBA	B1
24	--	--	V_{CC}	$\overline{B0}$
25	--	--	--	G
26	--	--	--	SBA
27	--	--	--	CBA
28	--	--	--	V_{CC}

FIGURE 1. Terminal connections.

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Device type 03

Input		Output
$\overline{OE}$	T/$\overline{R}$	
L	L	Bus B data to bus A
L	H	Bus A data to bus B
H	X	High Z state

Device type 05

Output enable input		Operation
$\overline{OEBA}$	OEBA	
L	L	B data to A bus
H	H	A data to B bus
H	L	Isolation

Device type 06

$\overline{OE}$	T/$\overline{R}$	Applied inputs	Valid direction I/P - O/P	Output
H	X	X	X	X
L	H	H	A to B	L
L	H	L	A to B	H
L	L	H	B to A	L
L	L	L	B to A	H

Device type 07

$\overline{OE}$	T/$\overline{R}$	Applied inputs	Valid direction I/P - O/P	Output
H	X	X	X	X
L	H	H	A to B	L
L	H	L	A to B	H
L	L	H	B to A	H
L	L	L	B to A	L

Device types 08 and 09

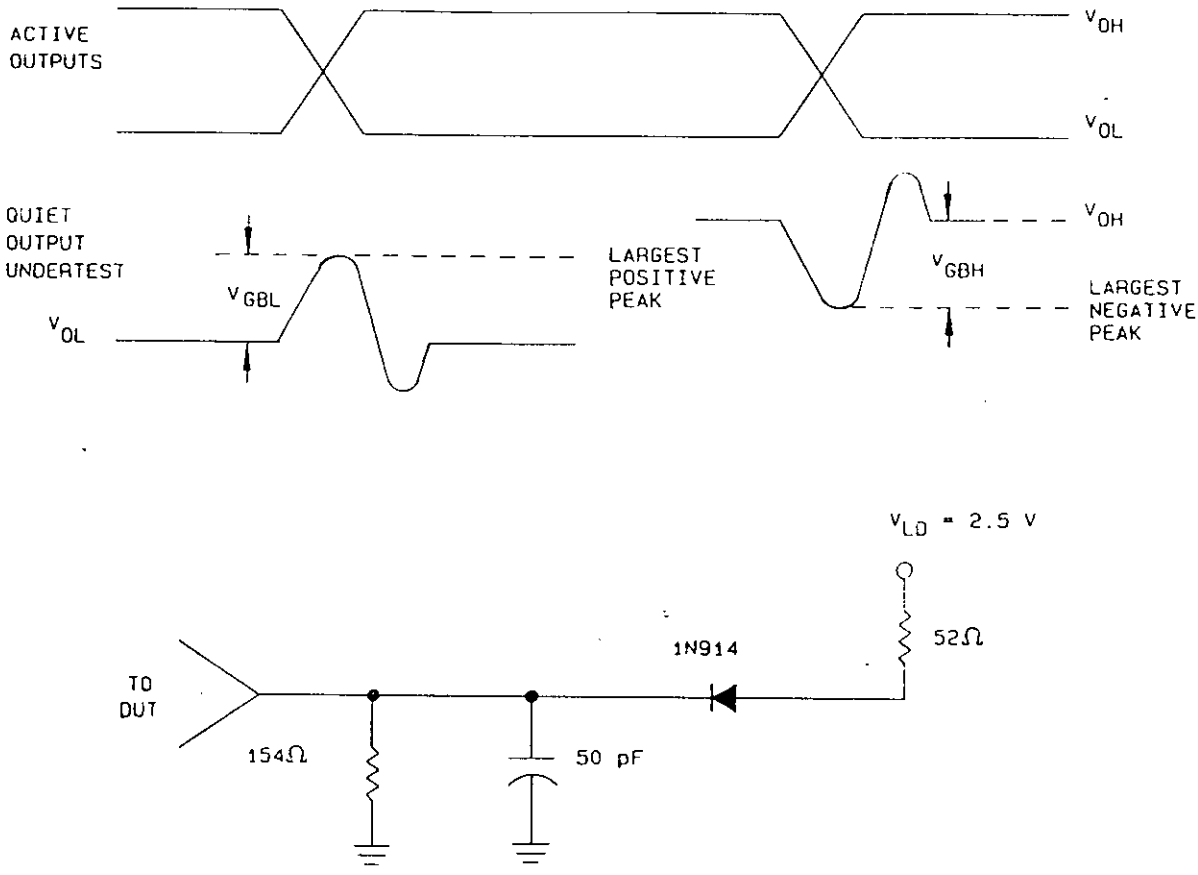
Inputs						Data I/O**		Operation
$\overline{G}$	DIR	CAB	CBA	SAB	SBA	A0-A7	B0-B7	
H	X	H or L	H or L	X	X	Input	Input	Isolation
H	X	↑	X	X	X			Store A and B data
H	X	X	↑	X	X			Store B and A data
L	H	X	X	L	X	Input	Output	Real time A* data to B bus
L	H	↑	X	L	X			Clock A* data into A register
L	H	H or L	X	H	X			A register* data to B bus
L	H	↑	X	H	X			Clock A* data into A register and output to B bus
L	L	X	X	X	L	Output	Input	Real time B* data to A bus
L	L	X	↑	X	L			Clock B* data into B register
L	L	X	H or L	X	H			B register* data to A bus
L	L	X	↑	X	H			Clock B* data into B register and output to A bus

* For device type 09, these data inputs, are inverted.
** The data output functions may be enabled or disabled by various signals at the $\overline{G}$ and DIR inputs. Data input functions are always enabled; i. e., data at the bus pins will be stored on every low-to-high transition of the clock inputs.

H = High voltage level
L = Low voltage level
X = Irrelevant
Z = High impedance
↑ = Low-to-High clock transition

FIGURE 2. Truth tables.

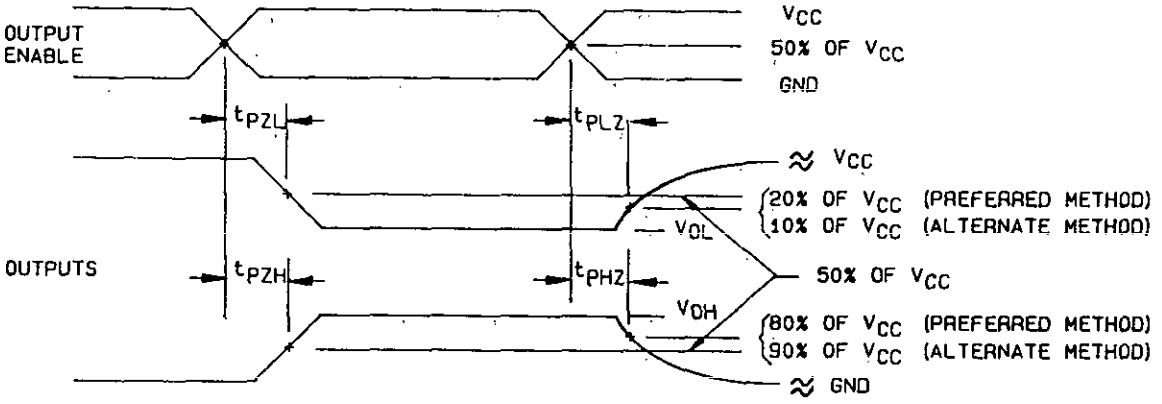
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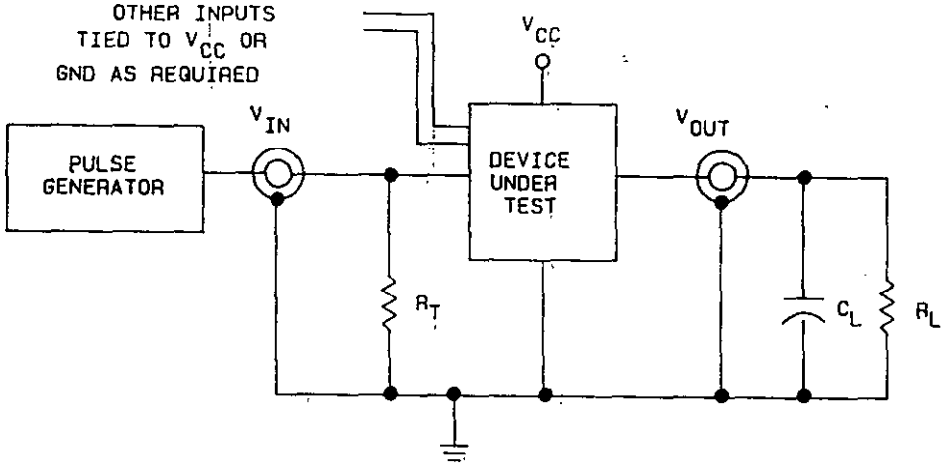


NOTE: Resistor and capacitor tolerance = ±10 percent

FIGURE 3. Voltage levels for ground bounce.

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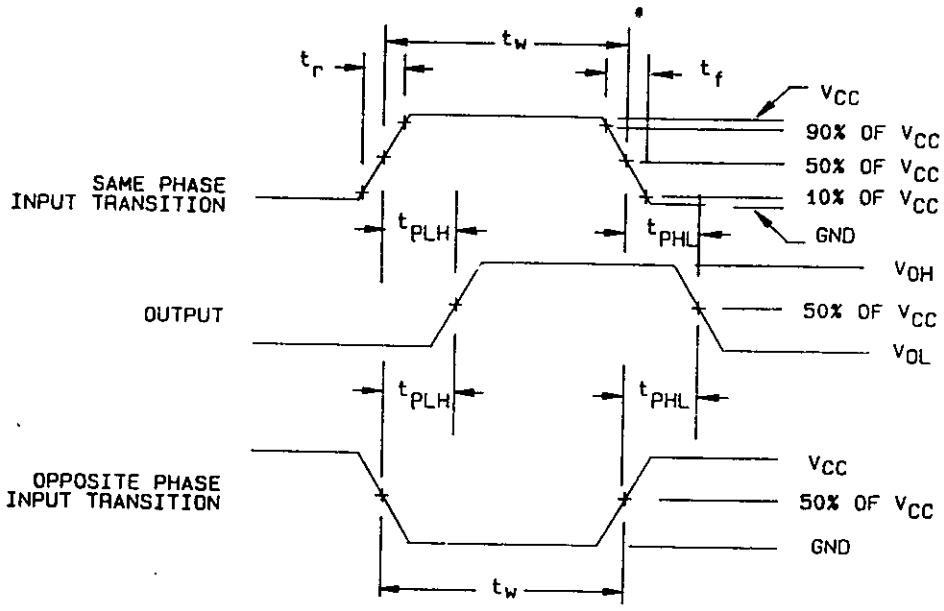


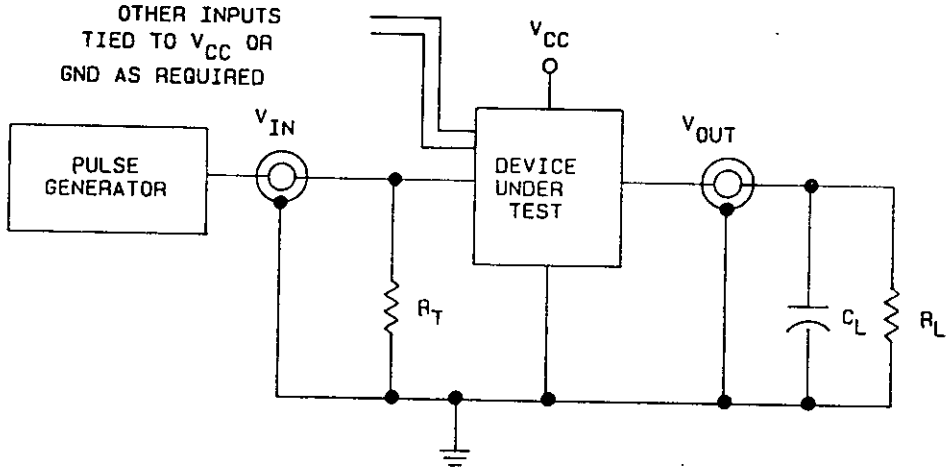


NOTES:
1. Referred method:
   when measuring t_{PHZ} or t_{PZH}: V_{TEST} = GND,
   when measuring t_{PLZ} or t_{PZL}: V_{TEST} = 2(V_{CC}).
2. Alternate method:
   when measuring t_{PHZ} or t_{PZH}: V_{TEST} = open,
   when measuring t_{PLZ} or t_{PZL}: V_{TEST} = 2(V_{CC}).
3. C_L = 50 pF or equivalent (includes test jig and probe capacitance).
4. R_L = 500Ω or equivalent.
5. R_T = 50Ω or equivalent.
6. V_{IN} = 0 V to V_{CC}.

FIGURE 4. Switching time waveforms and test circuits.

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NOTES:
1. $t_r \leq 3$ ns; $t_f \leq 3$ ns; PRR $\leq$ 10 MHz; duty cycle = 50 percent. For synchronous and asynchronous inputs, use as a maximum, the minimum pulse width or recovery time listed in Table I herein, as applicable, in place of the duty cycle.
2. C_L = 50 pF or equivalent (includes test jig and probe capacitance).
3. R_L = 500Ω or equivalent.
4. R_T = 50Ω or equivalent.

FIGURE 4. Switching time waveforms and test circuits - Continued.

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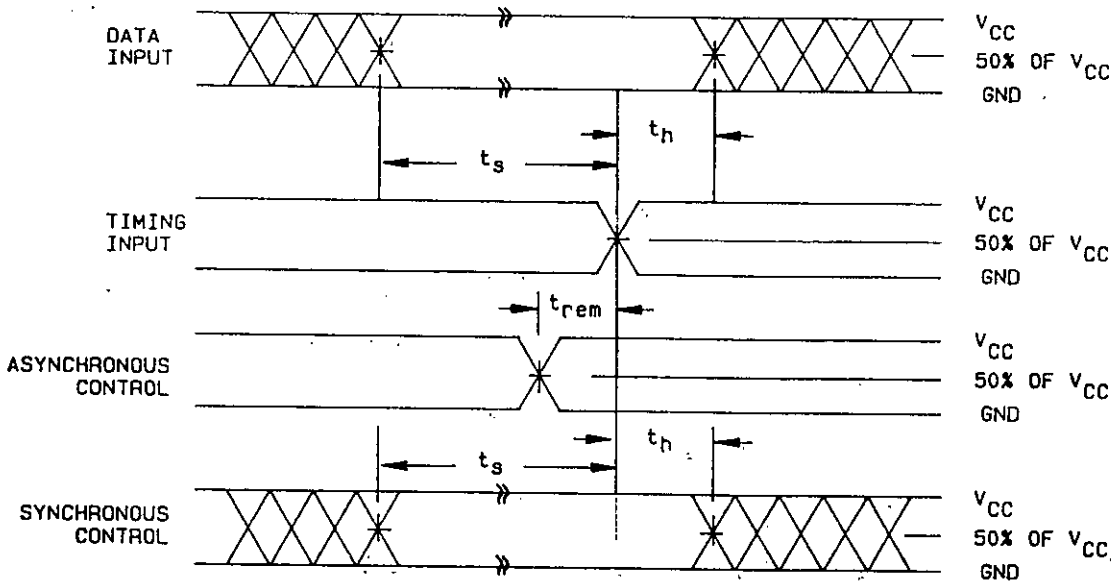


FIGURE 4. Switching time waveforms and test circuits - Continued.

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4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified in 3.1.10 through 3.1.14 of method 5004 and substitute lines 1 through 7 of table II herein.

b. Static burn-in, test condition A method 1015 of MIL-STD-883. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

(1) For static burn-in I, the output enable control pin(s) shall be connected to the resistors in parallel to V_{CC} or GND, as applicable, to enable the outputs. The directional control pin(s) shall be connected to either V_{CC} or GND. All other inputs shall be connected to GND. Outputs may be open or connected to $V_{CC}/2$. Resistors R1 are optional on both inputs and open outputs, and required on outputs connected to $V_{CC}/2$ ±0.5 V. R1 = 220Ω to 47 kΩ.

(2) For static burn-in II, the output enable control pin(s) shall be connected to the resistors in parallel to V_{CC} or GND, as applicable, to enable the outputs. The directional control pin(s) shall be connected to either V_{CC} or GND. All other inputs shall be connected through the R1 resistors to V_{CC}. Outputs may be open or connected to $V_{CC}/2$ ±0.5 V. Resistors are optional on open outputs, and required on outputs connected to $V_{CC}/2$ ±0.5 V. R1 = 220Ω to 47 kΩ.

(3) V_{CC} = 5.5 V ±0.5 V.

c. Dynamic burn-in, test condition D method 1015 of MIL-STD-883.

(1) Input resistors = 220Ω to 2 kΩ ±20 percent.

(2) Output resistors = 220Ω ±20 percent.

(3) V_{CC} = 5.5 V ±0.5 V.

(4) The output enable control pin(s) shall be connected to the resistors in parallel to V_{CC} or GND, as applicable, to enable the outputs. The directional control pin(s) shall be connected to either V_{CC} or GND. All other inputs shall be connected to the resistors in parallel to a clock pulse (CP). Outputs shall be connected to $V_{CC}/2$ through the resistors.

(5) CP = 25 kHz to 1 MHz square wave; duty cycle = 50 percent ±15 percent; V_{IH} = 4.5 V to V_{CC}, V_{IL} = 0 V ±0.5 V; t_r, $t_f \leq 100$ ns.

d. Interim and final electrical parameters shall be as specified in table II herein.

e. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter requirements.

4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failures shall be cumulative for determining the PDA.

c. The PDA for class B devices shall be in accordance with MIL-M-38510 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics, after burn-in, exceed the specified delta (Δ) limits or electrical parameter limits specified in table I, subgroup 1, are defective and shall be removed from the lot. The verified number of failed devices times 100 percent, divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

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TABLE II. Burn-in and electrical test requirements.

Line no.	Applicable tests and MIL-STD-883 test method	Class S device 1/			Class B device 1/		
		Reference paragraph	Table I sub-groups 2/	Table III delta limits 3/	Reference paragraph	Table I sub-groups 2/	Table III delta limits 3/
1	Interim electrical parameters (method 5004)		1			1	
2	Static burn-in I (method 1015)	4.2b 4.5.2	Required 4/			Not required	
3	Same as line 1		1	Δ			
4	Static burn-in II (method 1015)	4.2b 4.5.2	Required 4/		4.2b 4.5.2	Required 5/	
5	Same as line 1	4.2d	1*	Δ	4.2d	1*	Δ
6	Dynamic burn-in (method 1015)	4.2c 4.5.2	Required 4/			Not required	
7	Same as line 1	4.2d	1	Δ			
8	Final electrical parameters (method 5004)		1*,2,7* 9			1*,2,7 9 5/	
9	Group A test requirements (method 5005)	4.4.1	1,2,3,4, 7,8,9,10, 11		4.4.1	1,2,3,4, 7,8,9,10, 11	
10	Group B end-point electrical parameters (method 5005)	4.4.2	1,2,3, 7,8,9, 10,11	Δ			
11	Group C end-point electrical parameters (method 5005)				4.4.3	1,2	Δ
12	Group D end-point electrical parameters (method 5005)	4.4.4	1,2,3		4.4.4	1,2	
13	Group E end-point electrical parameters (method 5005)	4.4.5	1,7,9		4.4.5	1,7,9	

1/ Blank spaces indicate tests are not applicable.
2/ * indicates PDA applies to subgroup (see 4.2.1).
3/ Δ indicates delta limit shall be required only on table I subgroup 1, where specified, and the delta values shall be computed with reference to the previous interim electrical parameters (line 1).
4/ On all class S lots, the device manufacturer shall maintain read-and-record data (as a minimum on disk) for burn-in electrical parameters (group A, subgroup 1), in accordance with test method 5004 of MIL-STD-883. For pre burn-in and interim electrical parameters the read-and-record requirements are for delta measurements only.
5/ The device manufacturer may at his option either complete subgroup 1 electrical parameter measurements, including delta measurements, within 96 hours after burn-in completion (removal of bias); or may complete subgroup electrical measurements without delta measurements within 24 hours after burn-in completion (removal of bias). When the manufacturer elects to perform the subgroup 1 electrical parameter measurements without delta measurements, there is no requirement to perform the pre-burn-in electrical test (first interim electrical parameter test in table II.)

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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510, and 4.3.1 herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E, inspections (see 4.4.1 through 4.4.5).

4.3.1 Electrostatic discharge sensitivity qualification inspection. Electrostatic discharge sensitivity (ESDS) testing shall be performed in accordance with MIL-STD-883, method 3015. Only those device types that pass ESDS testing at 2,000 volts or greater shall be considered as conforming to the requirements of this specification. ESDS testing shall be measured only for initial qualification and after process or design changes which may affect ESDS classification.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. O/V, O/I (latch-up) tests and $V_{GBL/H}$ (ground bounce) tests shall be measured only for initial qualification and after process or design changes which may affect the performance of the device. Latch-up test shall be considered destructive. Test all applicable pins on 5 devices with no failures.

c. C_{IN}, $C_{I/O}$ and C_{PD} shall be measured only for initial qualification and after process or design changes which may affect capacitance. Capacitance shall be measured between the designated terminal and GND at a frequency of 1 MHz. C_{PD} shall be tested in accordance with the latest revision of JEDEC Standard No. 20 and table I herein. Test all applicable pins on 5 devices with no failures.

d. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance.

e. Subgroups 7 and 8 tests shall be sufficient to verify truth table.

f. f_{MAX} shall be measured only for initial qualification and after process or design changes which may affect the device frequency. Test all applicable pins on 22 devices with zero failures.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady-state life (accelerated) shall be conducted using test condition D of method 1005 of MIL-STD-883 and the circuit described in 4.2c herein, or equivalent as approved by the qualifying activity.

b. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 5 of group B inspections and shall consist of tests specified in table III herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table III herein.

b. Steady-state life test (method 1005 of MIL-STD-883) shall be conducted using test condition D and as specified in 4.5.2 herein using a circuit as described in 4.2c herein, or equivalent as approved by the qualifying activity.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

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TABLE III. Delta limits at 25°C.

Parameter	Device type	Limits
I_{CCZ}, I_{CCH}, I_{CCL} 1/	All	±100 nA

1/ The above parameters shall be recorded before and after the required burn-in and life tests to determine deltas (Δ).

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardness assured (see 3.5.1). RHA testing shall be performed in accordance with table V, method 5005 of MIL-STD-883 and electrical tests in table I herein. RHA quality conformance inspection sample tests shall be performed at the level specified in the purchase order.

a. RHA test for levels M, D, and R shall be performed through each level to determine at what levels the devices meet the RHA requirements. These RHA tests shall be performed for initial qualification and after design or process changes which may affect the RHA performance of the device.

b. End-point electrical parameters shall be as specified in table II herein.

c. Prior to total dose irradiation, each selected sample shall be assembled in its qualified package. It shall pass the specified group A electrical parameters for subgroups specified in table II herein.

d. The devices shall be subjected to radiation hardness assurance tests as specified in MIL-M-38510 for the RHA level being tested, and meet the postirradiation end-point electrical parameter limits as defined in table I at T_A = 25°C ±5°C, after exposure.

e. Prior to and during total dose irradiation characterization and testing, the devices for characterization shall be biased so that 50 percent of the devices are tested at input(s) high and 50 percent are at input(s) low, and the devices for testing shall be biased to the worst case condition established during characterization. Devices shall be biased as follows:

1. Inputs tested high, V_{CC} = 5.5 V dc +5 percent, R_{CC} = 10Ω ±20 percent, V_{IN} = 5.0 V dc +5 percent, R_{IN} = 1 kΩ ±20 percent, and all outputs are open. The output enable control pin(s) shall be connected to R_{IN} in parallel to V_{CC} or GND, as applicable, to enable the outputs.

2. Inputs tested low, V_{CC} = 5.5 V dc +5 percent, R_{CC} = 10Ω ±20 percent, V_{IN} = 0.0 V dc, R_{IN} = 1 kΩ ±20 percent, and all outputs are open. The output enable control pin(s) shall be connected to R_{IN} in parallel to V_{CC} or GND, as applicable, to enable the outputs.

f. Subgroups 1 and 2 of table V (method 5005) of MIL-STD-883 shall be tested as appropriate for device construction.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. Unless otherwise specified, all voltages given are referenced to the microcircuit GND terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When the burn-in and life tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to within 10°C of their power stable condition at room temperature; then, electrical parameter end-point measurements shall be performed.

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4.5.3 Quiescent supply current. When performing quiescent supply current measurements (I_{CC}), the meter shall be placed so that all currents flow through the meter.

4.6 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, shall be supplied.

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, RHA tests and steady-state life tests (see 3.4).

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.4).

e. Final electrical parameters data (see 4.2d).

f. RHA delta limits.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logistic support of existing equipment.

6.2 Acquisition requirements. The acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

c. Complete PIN.

d. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

e. Requirements for certificate of compliance, if applicable.

f. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

g. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

h. Requirements for product assurance and radiation hardness assurance options.

i. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the PIN. Unless otherwise specified, these requirements shall not apply to direct purchase by, or direct shipment to the Government.

j. Requirements for "JAN" marking.

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6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

C_{IN}	Input terminal-to-GND capacitance
GND	Ground zero voltage potential
I_{CC}	Quiescent supply current
I_{IL}	Input current low
I_{IH}	Input current high
T_C	Case temperature
T_A	Ambient temperature
V_{CC}	Positive supply voltage
C_{PD}	Power dissipation capacitance
V_{IC}	Positive input clamp voltage
V_{GB}	Ground bounce voltage
O/V	Latch-up over-voltage
O/I	Latch-up over-current
t_w	Trigger duration (width)
$C_{I/O}$	Input/output terminal to GND capacitance

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead finish C (see 3.3). Longer length leads and lead forming shall not affect the PIN.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges, postirradiation performance or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
*01	54AC242
*02	54AC243
03	54AC245
*04	54AC620
05	54AC623
06	54AC640
07	54AC643
08	54AC646
09	54AC648

* To be included at a later date.

6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static build up. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Avoid use of plastic, rubber, or silk in MOS areas.

e. Maintain relative humidity above 50 percent, if practical.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OC, SH

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1277)

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